



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
UNIT TEST
16-08-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

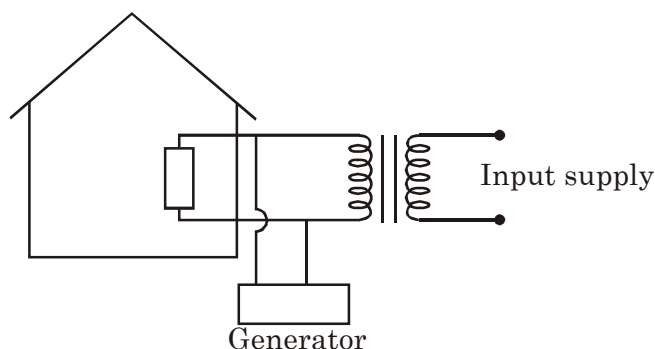
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A person connects a 120V AC generator to the household supply when power supply is off. The transformer is a step down transformer of turns ratios 100 : 1. What will be the voltage between the terminals of input supply experience by a repairman.



(A) 12,000 V

(B) 1.2 V

(C) 120 V

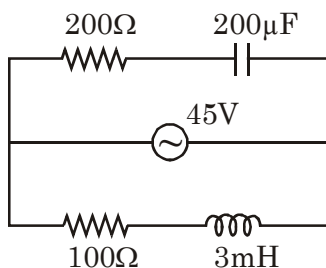
(D) 0

Space for Rough Work

2. Consider a room with 2 loudspeakers emitting sinusoidal sound waves which are coherent and in the same phase. A student moving from rear to front end hears sound getting repeatedly loud & soft but with same pitch. Choose the correct explanation for the phenomenon.



- (A) Student is a moving observer so she observes different frequencies for the sound. She is observing beats due to different doppler shifted frequencies.
 (B) The sound waves are forming a standing wave pattern in the room. The student is observing nodes and antinodes due to which she perceives varying intensity.
 (C) Both (A) & (B) are correct.
 (D) Neither (A) and nor (B) are correct explanations.
3. If the current delivered by source when frequency is very small is i_1 and current delivered by source when frequency is very large is i_2 , find i_1/i_2 .



(A) 1

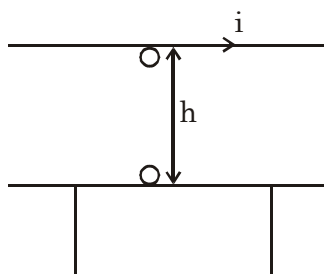
(B) 2

(C) $1/2$

(D) $1/4$

Space for Rough Work

4. A small circular washer of radius 0.5 cm is held below a long straight wire carrying current of 10A. It is located at $h = 0.5\text{m}$ above a table top. It is now dropped from rest. Assume that magnetic field over the washer is nearly constant and is equal to the value at its centre. What is the induced emf in the washer when it is just about to hit the table? The resistance of washer is very high.



(A) 2nV

(B) 1nV

(C) 3nV

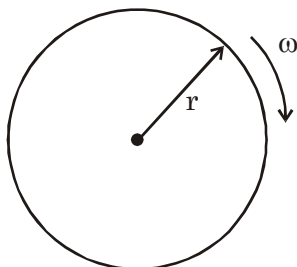
(D) 4nV

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. A rope of linear mass density μ and radius r is rotated on a smooth horizontal surface with an angular velocity ω . Sine waves can be generated in it which move in both direction. They superpose over themselves to produce a standing wave pattern. Which of the following can be one of its frequencies ?



(A) $\frac{\omega}{4\pi}$

(B) $\frac{\omega}{2\pi}$

(C) $\frac{\omega}{3\pi}$

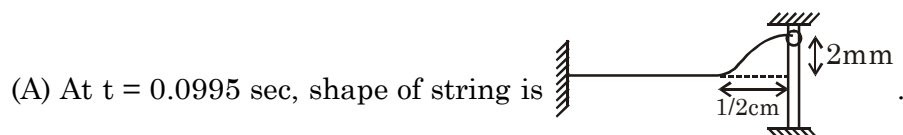
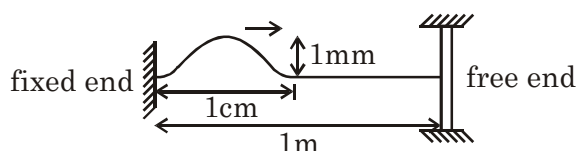
(D) $\frac{\omega}{\pi}$

Space for Rough Work

6. Three sinusoidal waves travel in +ve x-direction along the same string. All 3 waves have same frequency. Their amplitudes are A , $\frac{3A}{5}$ and $\frac{A}{5}$. Then phase angles are 0 , $\pi/2$ and π respectively. The resultant waveform can be :

(A) $y = A \sin\left(\omega t - kx + \frac{37\pi}{180}\right)$ (B) $y = A \sin\left(kx - \omega t + \frac{37\pi}{180}\right)$
 (C) $y = \frac{4A}{5} \sin\left(\omega t - kx + \frac{53\pi}{180}\right)$ (D) $y = A \sin\left(\omega t - kx + \frac{53\pi}{180}\right)$

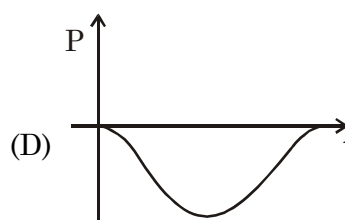
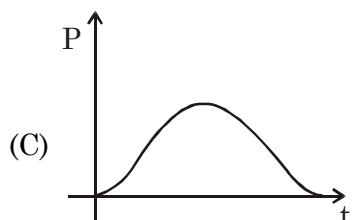
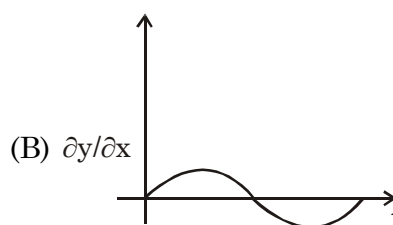
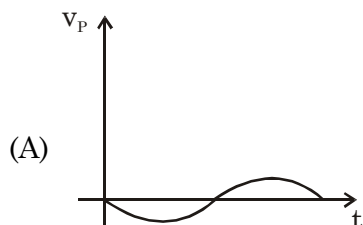
7. A pulse of shape $y = A \sin^2 \pi x$ was made on a string as shown. Tension in the string is 1N and mass/length of the string is 10 gm/m . The shape shown here is at $t = 0$. The pulse moves towards right. Choose the correct statements (s)



- (B) The string is completely straight after 0.1995 sec .
 (C) The string acquires its original shape at $t = 2.0\text{ sec}$
 (D) At $t = 0$, the potential energy of pulse and kinetic energy are equal.

Space for Rough Work

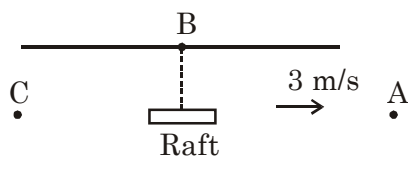
8. A boy standing at $x = 0$ holds a string taut. He moves his hand up & down to produce a pulse of shaped $y = (1\text{mm}) \sin^2\left(\frac{\pi}{2}x\right)$; $0 \leq x \leq 1\text{ cm}$ (x is in cm). This pulse travels in +ve x direction with a speed of 1 cm/s . If v_p = velocity of particle & $\frac{\partial y}{\partial x}$ = slope of string & P = power being transmitted. Choose appropriate graph(s) at $x = 1\text{cm}$.



9. Consider a sound wave whose equation is $S_1 = S_0 \sin(\omega t - kx)$ where symbols have usual meaning. Second wave has equation $S_2 = S_0 \sin(\omega t + kx)$. Choose the correct statement(s)
- (A) In the first wave; velocity of particles and excess pressure are in same phase.
 (B) In the first wave; velocity of particles and excess pressure are 180° out of phase.
 (C) In the second wave; velocity of particles and excess pressure are in same phase.
 (D) In the second wave; velocity of particles and excess pressure are 180° out of phase.

Space for Rough Work

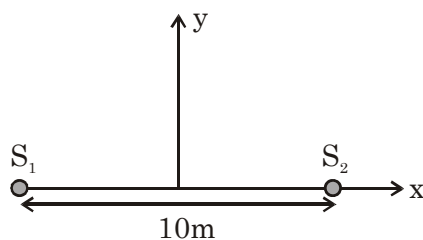
10. A raft is floating on a river. A boy sitting on the raft is making ripples with a frequency of 10 Hz. The ripples travel in water with speed of 5 m/s relative to water. A, B and C are stationary observers.



- (A) Frequency of ripples as seen by A is 16 Hz.
 (B) Frequency of ripples as seen by C is 6.25 Hz.
 (C) Frequency of ripples as seen by B is 10 Hz.
 (D) Frequency of ripples as seen by B is 25 Hz.
11. A pipe is fabricated in form of an organ pipe closed at both the ends. The gas enclosed is a mixture of H_2 and He (in ratio 1 : 1 by moles). One of the ends is fixed and the other end can be moved by a piston. The gas undergoes expansion to make its volume x times during a process for which molar heat capacity is constant ($C = R$). In the initial state, the gas is set into resonance by a tuning fork of frequency 100 Hz. In the final state, the gas is set into resonance by the same fork. The initial state is fundamental mode.
- (A) If final state is 8th harmonic, $x = 4$. (B) If final state is 3rd overtone, $x = 2$
 (C) If final state is 3rd harmonic, $x = 3$ (D) If final state is 26th overtone, $x = 9$

Space for Rough Work

12. In air where speed of sound is 344 m/s, two identical loudspeakers 10 m apart are driven by a same oscillator of frequency 21.5 Hz. Origin is at the mid point of line joining the sources S_1 & S_2 . At which of the following coordinates does the observer record a minima ? (All distance are in m)



- (A) (4, 0) (B) $\left(\frac{4\sqrt{10}}{3}, 1\right)$ (C) (5, 2.25) (D) $(3, 4\sqrt{3})$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

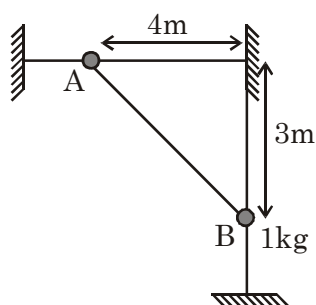
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+	●				
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

●	●	●	0	0	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

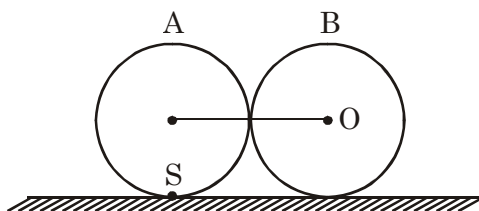
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Zero Marks : 0 In all other cases.

1. A 1.3m long rope has mass of 273gm. It has a tension of 21N. What must be the frequency (in Hz.) of travelling waves of amplitude 5 mm so that it transmits 378 W power on an average?
2. Two small beads of mass 1kg each are held at rest by holding bead A. The rods are smooth. A pulse is produced in the light string at A and it reaches B in time of $T/100$ sec. Now A was moved a distance of 1m to right and again held fixed. Again a pulse is produced in the string at A and it reached B in time of $\frac{T+1}{100}$ sec. What is mass of the light string (in gm)?



Space for Rough Work

3. A 10Ω resistor, 10 mH inductor and a $100\mu\text{F}$ capacitor are connected in series to a 50 V source having variable frequency. Find the energy delivered (in mJ) to the circuit during one time period if the operating frequency is twice the resonance frequency?
4. Two open organ pipes are simultaneously sounded together, Both of them have the same length and both are vibrated in fundamental mode. If velocity of sound is 320 m/s and we hear a beat frequency of 1Hz, what is the difference in their radii (in cm)? The length is 1m each. Assume that r_1 & $r_2 \ll 1\text{m}$.
5. Two vibrating tuning forks produce progressive waves given by $y_1 = 4\sin(500\pi t)$ and $y_2 = 2\sin(506\pi t)$. These tuning forks are held near the ear of a person. The person will hear α beats/s with intensity ratio between maxima and minima equal to β . Find the value of $\beta - \alpha$.
6. Two identical smooth dishes A, B of radius R hinged by a rigid rod of length $2R$ at their centres, are rolling on a horizontal surface with speed v_0 . A source S emitting a sound of frequency f is located on the periphery of disc A and a receiver is located on the centre of disc-B. Find the bandwidth of frequency recorded by receiver. Speed of sound in air is V. Take $V = 320\text{ m/s}$, $v_0 = 20\text{ m/s}$, $f = 510\text{ Hz}$.



Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 12)

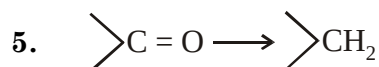
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- How many Faraday of charge is required to convert $\text{CH}_3\text{-NO}_2$ into 9.4 gm $\text{CH}_3\text{-NH(OH)}$
 (A) 0.1 F (B) 0.2 F (C) 0.4 F (D) 0.8 F
- A 500 ml solution is 0.1 M each in A^{+2} & B^{+2} , to this solution 500 ml of buffer solution having pH = 9 is added. What would be the concentration ratio $[\text{A}^{+2}] : [\text{B}^{+2}]$ finally.
 Given : $(K_{sp})_{\text{A(OH)}_2} = 4 \times 10^{-18} \text{ M}^3$, $(K_{sp})_{\text{B(OH)}_2} = 4 \times 10^{-10} \text{ M}^3$
 (A) 4 (B) 4×10^{-7} (C) 8×10^{-7} (D) 1
- Which of the following reagent can distinguish between SO_3^{2-} solution and SO_4^{2-} solution.
 (A) BaCl_2 solution (B) Pb(OAc)_2 solution
 (C) $\text{H}_2\text{S}_{(g)}$ (D) SrCl_2 solution
- $\text{CH}_3\text{-Br} \xrightarrow[\text{DMF}]{\text{KCN}} \text{X} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) PhMgCl}} \text{Y} \xrightarrow{\text{LiAlH}_4} \text{Z}$
 Z is :
 (A) $\text{Ph-CH}_2\text{-OH}$ (B) $\text{Ph}-\underset{\text{OH}}{\text{CH}}-\text{Ph}$ (C) $\text{Ph-CH}_2\text{-CH}_2\text{-OH}$ (D) $\text{Ph}-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

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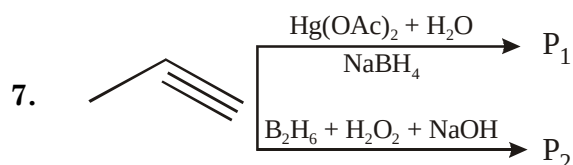
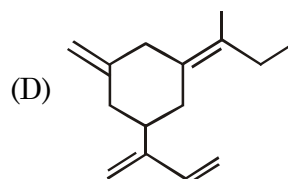
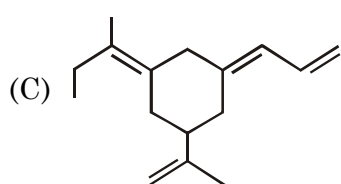
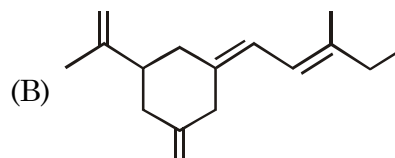
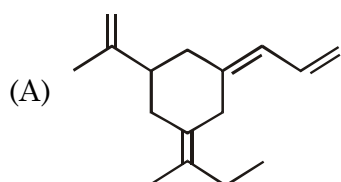
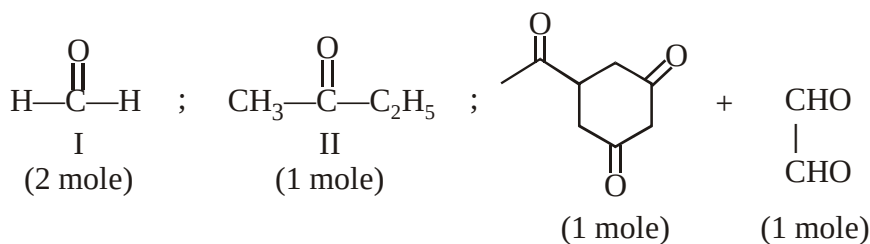


Reagent used for this conversion can be

- (A) Zn(Hg) + HCl (B) LiAlH_4 (C) Red P + HI (D) NaBH_4

Space for Rough Work

6. By which compound's reductive ozonolysis the following products are obtained



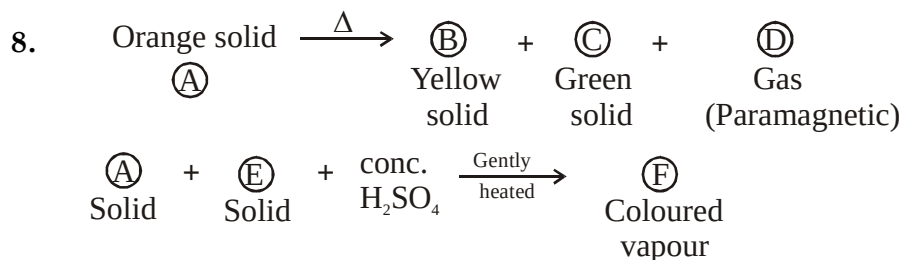
P_1 & P_2 will be distinguished by

(A) 2, 4, D.N.P

(B) alkaline Cu^{2+} (tartarate)

(C) NaHSO_3

(D) KMnO_4



The solid (E) may be :

(A) NH_4Cl

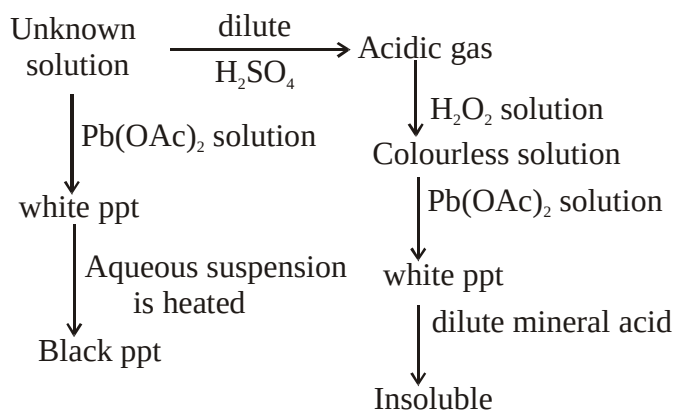
(B) NaBr

(C) KI

(D) Hg_2Cl_2

Space for Rough Work

9. For the given aqueous reactions, which of the statement(s) is(are) true?



- (A) Acidic gas is H_2S
- (B) Black ppt is PbS
- (C) Sodium salt of **acidic gas** react with $(\text{Zn} + \text{dilute } \text{H}_2\text{SO}_4)$ gives colourless gas having rotten egg smell
- (D) Dilute HNO_3 can dissolve black ppt.

10. Borax bead test is/are applicable for :

- (A) NaCl (B) CrCl_3 (C) CoCl_2 (D) KCl

Space for Rough Work

11. Select the correct statements(s)
- (A) Nernst equation is applicable only for reversible half cell or reversible net cell reaction but not applicable for irreversible conditions
 - (B) Gel in salt bridge permits diffusion of ions but eliminates convection current
 - (C) In galvanic cell overall cell reaction not necessarily an oxidation-reduction reaction
 - (D) The standard emf of a galvanic cell is the limiting value of E_{cell} as concentration tends to zero.
12. $\text{Zn} \left| \text{Zn}^{2+} \right|_{(0.10\text{M}, 1\text{l})} \parallel \text{Cu}^{2+} \left| \text{Cu(s)} \right|_{(0.9\text{M}, 1\text{l})}$ is used in electro-refining of silver (using impure 'Ag' anodes & pure 'Ag' cathode) at 25°C. When E_{cell} decreases to E_{cell}^0 then select the correct statement(s) at this stage
- (A) Loss in anodic mass of electrolytic cell is greater than 86.4 gm
 - (B) Loss in anodic mass of electrolytic cell is exactly 86.4 gm
 - (C) Increase in mass of 'Ag' cathode is 86.4 gm
 - (D) Increase in mass of 'Ag' cathode is 8.64 gm

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

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For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

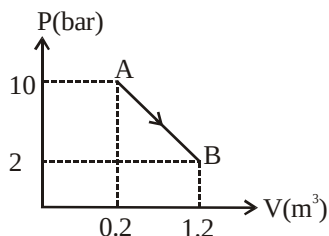
-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- For any vander waal gas $Z = 1 - \frac{0.105l}{V_m} + \frac{4 \times 10^{-4} l^2}{V_m^2} + \dots$ at 400K. $\frac{a}{b}$ for gas in $\frac{\text{atm} \cdot l}{\text{mol}}$ is.

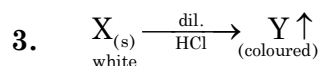
[R = 0.08 L-atm/mol-K]

- 10 mol of real gas X^{150} undergoes following expansion A to B



If it's molar internal energy 'U' is given as $U = (0.225 PV - 0.75)$ then calculate 'q' for the process in kJ.

Space for Rough Work



Compound 'X' gives golden yellow colour to the flame.

If compound 'X' has been allowed to react with following species in aqueous solution :

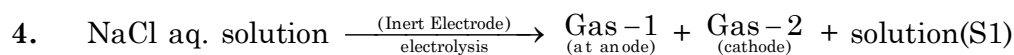
- (a) KMnO_4/H^+ (b) AgNO_3 (c) BaCl_2 (d) $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$
 (e) $\text{Pb}(\text{OAc})_2$ (f) $(\text{I}^- + \text{starch})$

Then calculate the value of $\left(\frac{Q}{P \times R} \right)$ where

P : Number of species which undergoes redox reaction with compound 'X'

Q : Number of species which gives ppt with compound 'X'

R : Number of species with which compound 'X' produce no observable change



Gas-1 + KI aq. $\longrightarrow$ diatomic compound (X) + solution (S2)

Solution (S1) + salt (Y) $\longrightarrow$ Gas-3 + solution (S-3)

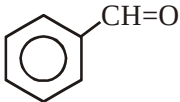
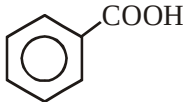
Salt-(Y) + conc. H_2SO_4 $\longrightarrow$ Gas-4 + other products

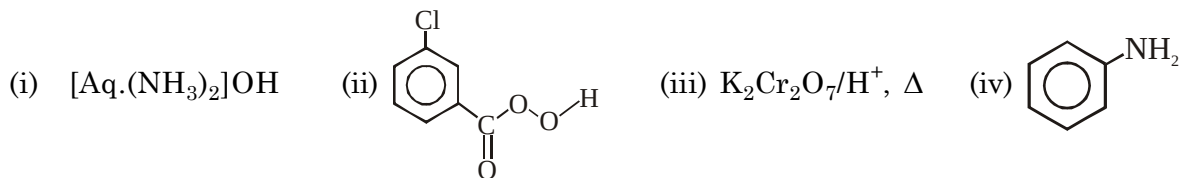
Gas-3 + Gas-4 $\longrightarrow$ white fumes of Salt-Y

Find sum of the molecular weight of Gas-1 + Gas-4 + diatomic compound (X)

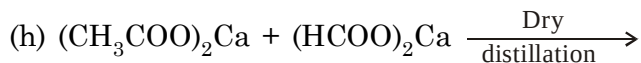
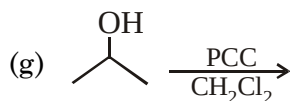
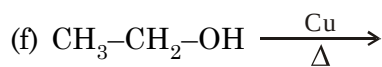
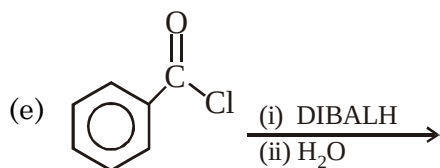
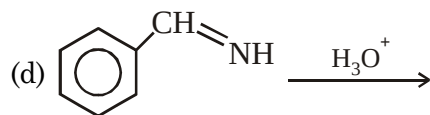
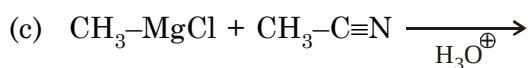
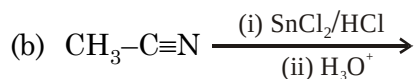
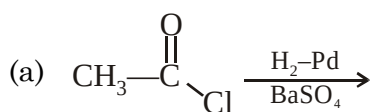
Given atomic weight of H = 1 , Cl = 35.5 , N = 14, I = 127 , Na = 23, K = 39, S = 32, O = 16

Space for Rough Work

5. Number of reagents by which  can be converted in to  followed by acidification.



6. Number of reactions in which aldehyde is obtained as one of the product.



Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The degree of the differential equation satisfying $\sqrt{1+x^2} + \sqrt{1+y^2} = A(x\sqrt{1+y^2} - y\sqrt{1+x^2})$ is (where 'A' is real parameter)
 (A) 1 (B) 2 (C) 3 (D) 4
2. Given 'P' be any arbitrary interior point of the triangle ABC and P_a, P_b, P_c be the distances of P from sides BC, CA, AB respectively. If h_a, h_b, h_c be the length of altitudes drawn from vertices A, B and C respectively to the opposite sides, then $\frac{P_a}{h_a} + \frac{P_b}{h_b} + \frac{P_c}{h_c}$ is equal to-
 (A) 1 (B) 2 (C) 3 (D) $\frac{\text{Inradius}}{\text{Circumradius}}$
3. If A and B are two idempotent matrices of same order and $A + B$ is idempotent, then -
 (A) $AB + BA = I$ (B) $AB = O$ (C) $A = B$ (D) $AB = I$
4. The integral value of 'n' for which $\lim_{x \rightarrow 0} \frac{\int_0^x (\cos 2t - 1)(\cos t - e^{-t})t^{-n} dt}{\cos x - 1}$ is a finite non-zero number, is
 (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Let $f(x) = \int_1^x \frac{3^t}{1+t^2} dt$, where $x > 0$, then which is true ?
 (A) for $0 < \alpha < \beta$, $f(\alpha) < f(\beta)$ (B) for $0 < \alpha < \beta$, $f'(\alpha) < f'(\beta)$
 (C) $f(x) + \frac{\pi}{4} < \tan^{-1} x \forall x \geq 1$ (D) $f(x) + \frac{\pi}{4} > \tan^{-1} x \forall x \geq 1$
6. Let f be twice differentiable function defined in $[-3, 3]$ such that $f(0) = -4$, $f'(3) = 0$, $f'(-3) = 12$ and $f''(x) \geq -2 \forall x \in [-3, 3]$. If $g(x) = \int_0^x f(t) dt$ then which of the following is/are true ?
 (A) $f(x) = 0$ has one positive root and one negative root
 (B) $g(3) = 6$
 (C) maximum value of $f(x)$ in $[-3, 3]$ is 5.
 (D) length of latus rectum of $y = f(x)$ is 1 unit

Space for Rough Work

7. Let α be the region in the plane $x + y + z = 1$ with $x, y, z > 0$. β is region consisting set of points (a, b, c) in α such that only two of the three inequalities : $a \leq \frac{1}{2}, b \leq \frac{1}{3}, c \leq \frac{1}{6}$ hold simultaneously.

If area of region R is denoted by $A(R)$, then

(A) $A(\alpha) = \frac{\sqrt{3}}{4}$ (B) $A(\alpha) = \frac{\sqrt{3}}{2}$ (C) $A(\beta) = \frac{7\sqrt{3}}{36}$ (D) $A(\beta) = \frac{11\sqrt{3}}{36}$

8. Let $f(x)$ be a function given by $f(x) = x^{3/2}e^{-x/2}$. Then -

(A) f is non-monotonic in $(4, \infty)$ (B) $\frac{9\sqrt{3}}{e\sqrt{e}} < \int_0^3 f(x)dx < \frac{9\sqrt{3}}{\sqrt{e}}$

(C) $\int_0^3 f(x)dx < \frac{9\sqrt{3}}{e\sqrt{e}}$ (D) f is monotonic in $(4, \infty)$

9. Identify the correct statement(s)

- (A) For two non-singular matrices A, B of same order, if $\text{adj}(A) = \text{adj}(B)$, then $A = B$
 (B) For two non-singular matrices A, B of same order, if $\text{adj}(A) = \text{adj}(B)$, then $|A| = |B|$
 (C) For two non-singular matrices A, B of same order, if $\text{adj}(A) = \text{adj}(B)$, then $A + B = 0$ or $A - B = 0$
 (D) For two non-singular matrices A, B of same order, if $\text{adj}(A) = \text{adj}(B)$, then $|A + B| = 0$ or $|A - B| = 0$
 (where $\text{adj}(X)$ denotes adjoint of matrix X and $|X|$ denotes value of its determinant)

Space for Rough Work

10. Which of the following statement is/are correct about two square matrices A and B of same order ?
 (A) $\text{trace}((A - 2B)(2A + 2B)) = 2\text{trace}(A^2) - 4\text{trace}(B^2)$
 (B) $(\text{Adj} A)(\text{Adj} A - 2I) = (\text{Adj} A - I)^2 - I$
 (C) $(\text{Adj} A)(\text{Adj} A - 2I) \neq (\text{Adj} A - I)^2 - I$
 (D) If A and B are symmetric matrices, then $((AB^2)^2)' = (B^2A)^2$
 (where $\text{Adj} A = \text{Adjoint of } A$, $\text{trace}(A) = \text{sum of diagonal elements of } A$)
11. A plane passing through (1, 1, 1) cut positive direction of co-ordinate axes at A, B and C, then the volume V of tetrahedron OABC can be (in cubic units)
 (A) 4 (B) $\frac{9}{2}$ (C) 5 (D) $\frac{11}{2}$
12. If the area bounded by the curves $x = y - y^2$ and $x = my$ is $\frac{9}{2}$ square units, then the value of m can be
 (A) $\frac{1}{2}$ (B) -2 (C) 3 (D) 4

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+	●				
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

●	●	●	0	0	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
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8	8	8	8	8	8
9	9	9	9	9	9

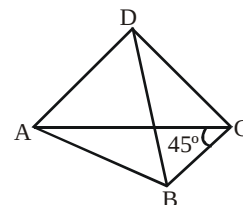
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If $\int \frac{\cos^5 x}{\sin x} dx = \ell \ln |\sin x| + k \sin^4 x - \sin^2 x + C$, (where C is constant of integration), then k is equal to

2. As shown in diagram, the volume of tetrahedron ABCD is $\frac{1}{6}$ cu. units.

Also $\angle ACB = 45^\circ$, $BC = 1$, $AD + BC + \frac{AC}{\sqrt{2}} = 3$, then length of

CD is equal to (in units)



Space for Rough Work

3. If the value of integral $\int_0^{\pi/4} \frac{\sin^2 7\theta}{\sin 2\theta} d\theta = \frac{2}{3} + k \ln 2$, then the value of k is equal to
4. If the lines $\frac{x-1}{2} = \frac{y}{-1} = \frac{z}{2}$ and $x - y + z - 2 = 0 = \lambda x + 3z + 5$ are coplanar, then λ is equal to
5. Let $L_1 : \vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda(\hat{i} + \hat{j})$, $\lambda \in \mathbb{R}$ and $L_2 : \vec{r} = \hat{i} + (\mu + 1)\hat{j} + (\mu + 1)\hat{k}$, $\mu \in \mathbb{R}$ be the two lines intersecting at point A. Through point B(3, 3, 1) a line is drawn making an angle of 60° with L_2 and intersecting it at point C, then area of ΔABC is equal to
6. Let $A = [a_{ij}]_{2 \times 2}$ such that $a_{ij} \in \{-1, 0, 1\} \forall i, j$ and $\text{adj}(A) = -A$, $\det(A) = -1$. Then number of such matrices A is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	12
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–1 In all other cases	32
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator